Name: _____ Register No: _____

Experiment 4

Question: To count the frequency of occurrence of a word in a body of text is often needed during text processing.

Code

```
import json
import re
from collections import Counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

def exp_4():
    text = ("Data science is fun. Data science uses Python, and Python makes "
            "data analysis easy. Science needs data!")
    words = re.findall(r"\b[a-z']+\b", text.lower())   # clean + tokenize

    # Method 1: dictionary
    freq = {}
    for w in words:
        freq[w] = freq.get(w, 0) + 1
    print("Using dict   :", freq)

    # Method 2: collections.Counter
    counter = Counter(words)
    print("Using Counter:", counter)
    print("Top 3        :", counter.most_common(3))

    # Method 3: pandas
    print(pd.Series(words).value_counts().head(5))


exp_4()
```

Output

```
Using dict   : {'data': 4, 'science': 3, 'is': 1, 'fun': 1, 'uses': 1, 'python': 2, 'and': 1,
    'makes': 1, 'analysis': 1, 'easy': 1, 'needs': 1}
Using Counter: Counter({'data': 4, 'science': 3, 'python': 2, 'is': 1, 'fun': 1, 'uses': 1, 'and':
     1, 'makes': 1, 'analysis': 1, 'easy': 1, 'needs': 1})
Top 3        : [('data', 4), ('science', 3), ('python', 2)]
data       4
science    3
python     2
is         1
fun        1
Name: count, dtype: int64
```